

# Advanced TA Session

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Expectation, Variance, Sample Mean, Sample  
Variance, Covariance,  
Correlation Coefficient, Jointly Gaussian Variables,  
Cauchy–Schwarz

*The first 10 pages contain one problem per page with space left for solving. Detailed stepwise solutions are provided after the problem set.*

**Problem 1**

Let  $X$  denote the number of clicks received by a webpage in one minute. Suppose

$$E[X] = 8, \quad \text{Var}(X) = 6.$$

Compute:

- (a)  $E[3X + 5]$
- (b)  $\text{Var}(3X + 5)$

**Problem 2**

Suppose  $X_1, X_2, \dots, X_n$  are i.i.d. random variables representing the prediction error of a machine learning model on  $n$  independent data points. Let

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

Show that

$$E[\bar{X}] = E[X_1]$$

and compute  $\text{Var}(\bar{X})$  in terms of  $\text{Var}(X_1)$ .

**Problem 3**

Let  $X$  denote the response time of a server request, with

$$E[X] = \mu, \quad \text{Var}(X) = \sigma^2.$$

Define

$$Y = X - \mu.$$

Compute

$$E[Y] \quad \text{and} \quad \text{Var}(Y).$$

Briefly interpret the result.

**Problem 4**

A data scientist evaluates a classifier on  $n$  independent samples. Define

$$X_i = \begin{cases} 1, & \text{if prediction } i \text{ is correct,} \\ 0, & \text{otherwise.} \end{cases}$$

Assume

$$P(X_i = 1) = p.$$

Let

$$\hat{p} = \frac{1}{n} \sum_{i=1}^n X_i$$

be the observed accuracy.

Find

$$E[\hat{p}] \quad \text{and} \quad \text{Var}(\hat{p}).$$

**Problem 5**

Suppose  $X$  and  $Y$  represent the number of clicks on two advertisements shown to the same user.

Starting from the definition

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])],$$

show that

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y].$$

**Problem 6**

A recommender system produces a predicted relevance score  $X$ , and the corresponding actual engagement score is  $Y$ . Suppose

$$\text{Var}(X) = 4, \quad \text{Var}(Y) = 9, \quad \text{Cov}(X, Y) = 3.$$

Compute the correlation coefficient

$$\rho_{XY}.$$

Briefly interpret the answer.

**Problem 7**

Suppose  $X$  and  $Y$  are jointly Gaussian random variables.  
Show that if

$$\text{Cov}(X, Y) = 0,$$

then  $X$  and  $Y$  are independent.

Also explain why this implication does not generally hold for arbitrary non-Gaussian random variables.



**Problem 8**

Let  $X_1, \dots, X_n$  be independent observations of the loss of a neural network on  $n$  training examples. Suppose the population variance is  $\sigma^2$ .

The estimator

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

is used to estimate  $\sigma^2$ .

Explain why dividing by  $n-1$  instead of  $n$  gives an unbiased estimator of the population variance.

**Problem 9**

Let  $X$  and  $Y$  be two random variables representing two features in a machine learning dataset.

Using the Cauchy–Schwarz inequality, prove that

$$|\text{Cov}(X, Y)| \leq \sqrt{\text{Var}(X)\text{Var}(Y)}.$$

Hence deduce that the correlation coefficient satisfies

$$-1 \leq \rho_{XY} \leq 1.$$

**Problem 10**

Suppose a dataset contains  $n$  independent observations of a random variable  $X$  with mean  $\mu$  and variance  $\sigma^2$ . Let

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

denote the sample mean.

Explain why increasing  $n$  reduces the variance of  $\bar{X}$ , and discuss the implications of this fact for training and evaluating machine learning models on larger datasets.

## Detailed Stepwise Solutions

### Solution to Problem 1

**Step 1: Use linearity of expectation.**

For any constants  $a, b$ ,

$$E[aX + b] = aE[X] + b.$$

Here  $a = 3$  and  $b = 5$ , so

$$E[3X + 5] = 3E[X] + 5.$$

**Step 2: Substitute the given expectation.**

$$E[3X + 5] = 3(8) + 5 = 24 + 5 = 29.$$

Thus

$$E[3X + 5] = 29.$$

**Step 3: Use the variance transformation rule.**

For constants  $a, b$ ,

$$\text{Var}(aX + b) = a^2 \text{Var}(X).$$

So

$$\text{Var}(3X + 5) = 3^2 \text{Var}(X) = 9 \cdot 6 = 54.$$

Hence

$$\text{Var}(3X + 5) = 54.$$

### Solution to Problem 2

**Step 1: Write the sample mean.**

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

**Step 2: Compute the expectation.**

Using linearity of expectation,

$$E[\bar{X}] = E\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n E[X_i].$$

Since the variables are i.i.d.,

$$E[X_i] = E[X_1] \quad \text{for all } i.$$

Therefore

$$E[\bar{X}] = \frac{1}{n} \sum_{i=1}^n E[X_1] = \frac{nE[X_1]}{n} = E[X_1].$$

So

$$E[\bar{X}] = E[X_1].$$

**Step 3: Compute the variance.**

Since

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i,$$

we have

$$\text{Var}(\bar{X}) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \text{Var}\left(\sum_{i=1}^n X_i\right).$$

Because the  $X_i$  are independent,

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i).$$

Again, since they are i.i.d.,

$$\text{Var}(X_i) = \text{Var}(X_1).$$

Thus

$$\text{Var}(\bar{X}) = \frac{1}{n^2} \cdot n \text{Var}(X_1) = \frac{\text{Var}(X_1)}{n}.$$

Hence

$$\boxed{\text{Var}(\bar{X}) = \frac{\text{Var}(X_1)}{n}}.$$

**Solution to Problem 3****Step 1: Write the transformed variable.**

$$Y = X - \mu.$$

**Step 2: Compute the expectation.**

Using linearity,

$$E[Y] = E[X - \mu] = E[X] - \mu.$$

But

$$E[X] = \mu,$$

so

$$E[Y] = \mu - \mu = 0.$$

Therefore

$$\boxed{E[Y] = 0}.$$

**Step 3: Compute the variance.**

Subtracting a constant does not change variance:

$$\text{Var}(Y) = \text{Var}(X - \mu) = \text{Var}(X) = \sigma^2.$$

Hence

$$\boxed{\text{Var}(Y) = \sigma^2}.$$

**Step 4: Interpretation.**

Centering a random variable by subtracting its mean makes the new variable have mean zero, but it does *not* change its spread. The variance remains the same.

**Solution to Problem 4****Step 1: Identify the distribution of  $X_i$ .**

Each  $X_i$  is Bernoulli with parameter  $p$ , so

$$E[X_i] = p, \quad \text{Var}(X_i) = p(1 - p).$$

**Step 2: Compute the expectation of  $\hat{p}$ .**

We have

$$\hat{p} = \frac{1}{n} \sum_{i=1}^n X_i.$$

Therefore,

$$E[\hat{p}] = \frac{1}{n} \sum_{i=1}^n E[X_i] = \frac{1}{n} \sum_{i=1}^n p = \frac{np}{n} = p.$$

So

$$E[\hat{p}] = p.$$

**Step 3: Compute the variance of  $\hat{p}$ .**

Since the  $X_i$  are independent,

$$\text{Var}(\hat{p}) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i).$$

Thus

$$\text{Var}(\hat{p}) = \frac{1}{n^2} \cdot np(1 - p) = \frac{p(1 - p)}{n}.$$

Hence

$$\text{Var}(\hat{p}) = \frac{p(1 - p)}{n}.$$

**Step 4: Interpretation.**

The estimator  $\hat{p}$  is unbiased, and its variance decreases as  $n$  increases. So larger test sets produce a more stable estimate of model accuracy.

**Solution to Problem 5****Step 1: Start from the definition.**

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])].$$

**Step 2: Expand the product inside the expectation.**

$$(X - E[X])(Y - E[Y]) = XY - XE[Y] - E[X]Y + E[X]E[Y].$$

So

$$\text{Cov}(X, Y) = E[XY - XE[Y] - E[X]Y + E[X]E[Y]].$$

**Step 3: Use linearity of expectation.**

$$\text{Cov}(X, Y) = E[XY] - E[XE[Y]] - E[E[X]Y] + E[E[X]E[Y]].$$

**Step 4: Pull out constants.**

Since  $E[X]$  and  $E[Y]$  are constants,

$$E[XE[Y]] = E[Y]E[X],$$

$$E[E[X]Y] = E[X]E[Y],$$

and

$$E[E[X]E[Y]] = E[X]E[Y].$$

Thus

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] - E[X]E[Y] + E[X]E[Y].$$

This simplifies to

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y].$$

Hence

$$\boxed{\text{Cov}(X, Y) = E[XY] - E[X]E[Y].}$$

**Solution to Problem 6****Step 1: Recall the definition of correlation coefficient.**

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}.$$

**Step 2: Substitute the given values.**

$$\rho_{XY} = \frac{3}{\sqrt{4 \cdot 9}}.$$

**Step 3: Simplify.**

$$\sqrt{4 \cdot 9} = \sqrt{36} = 6.$$

Therefore

$$\rho_{XY} = \frac{3}{6} = \frac{1}{2}.$$

So

$$\boxed{\rho_{XY} = \frac{1}{2}.}$$

**Step 4: Interpretation.**

A correlation of  $\frac{1}{2}$  indicates a moderate positive linear relationship: larger predicted relevance scores tend to be associated with larger engagement values.

**Solution to Problem 7****Step 1: State the special property of jointly Gaussian variables.**

For jointly Gaussian random variables, the joint distribution is completely determined by:

- the means,
- the variances,

- the covariance.

**Step 2: Use the condition**  $\text{Cov}(X, Y) = 0$ .

If

$$\text{Cov}(X, Y) = 0,$$

then the correlation is also zero.

For jointly Gaussian variables, zero covariance implies that the joint density factors into the product of the two marginal densities. Hence

$$f_{X,Y}(x, y) = f_X(x)f_Y(y),$$

which means  $X$  and  $Y$  are independent.

Therefore

If  $X, Y$  are jointly Gaussian and  $\text{Cov}(X, Y) = 0$ , then  $X$  and  $Y$  are independent.

**Step 3: Why this fails in general.**

For arbitrary random variables, zero covariance only means there is no *linear* dependence. It does not rule out nonlinear dependence.

For example, let  $X$  take values  $-1$  and  $1$  with equal probability, and let

$$Y = X^2.$$

Then  $Y = 1$  always, so  $Y$  depends completely on  $X$ , hence they are not independent.

But

$$\text{Cov}(X, Y) = 0.$$

So, in general,

$$\text{Cov}(X, Y) = 0 \not\Rightarrow X \text{ and } Y \text{ are independent}$$

unless we are in a special family such as the jointly Gaussian case.

## Solution to Problem 8

**Step 1: Recall the goal.**

We want an estimator of the population variance  $\sigma^2$  that satisfies

$$E[S^2] = \sigma^2.$$

**Step 2: Understand the issue with using  $\bar{X}$ .**

The quantity

$$\sum_{i=1}^n (X_i - \bar{X})^2$$

measures spread around the *sample mean*  $\bar{X}$ , not around the true mean  $\mu$ .

Because  $\bar{X}$  is estimated from the same data, it already adapts to the sample. This reduces the observed spread slightly.

**Step 3: Explain the loss of one degree of freedom.**



Once the sample mean  $\bar{X}$  is fixed, the deviations

$$(X_1 - \bar{X}), \dots, (X_n - \bar{X})$$

cannot vary independently. They must satisfy

$$\sum_{i=1}^n (X_i - \bar{X}) = 0.$$

So only  $n - 1$  of these deviations are free to vary.

That is why we say one degree of freedom is lost.

**Step 4: Consequence.**

If we divide by  $n$ , the resulting estimator is systematically too small:

$$E \left[ \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \right] = \frac{n-1}{n} \sigma^2.$$

To correct this bias, we divide by  $n - 1$  instead:

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Then

$$E[S^2] = \sigma^2.$$

Thus

Dividing by  $n - 1$  corrects the downward bias and makes  $S^2$  unbiased.

### Solution to Problem 9

**Step 1: Rewrite covariance in centered form.**

Let

$$U = X - E[X], \quad V = Y - E[Y].$$

Then

$$\begin{aligned} \text{Cov}(X, Y) &= E[UV], \\ \text{Var}(X) &= E[U^2], \quad \text{Var}(Y) = E[V^2]. \end{aligned}$$

**Step 2: Apply the Cauchy–Schwarz inequality.**

For random variables  $U$  and  $V$ ,

$$|E[UV]| \leq \sqrt{E[U^2]E[V^2]}.$$

Substituting the centered variables gives

$$|\text{Cov}(X, Y)| = |E[UV]| \leq \sqrt{E[U^2]E[V^2]} = \sqrt{\text{Var}(X)\text{Var}(Y)}.$$

Hence

$$|\text{Cov}(X, Y)| \leq \sqrt{\text{Var}(X)\text{Var}(Y)}.$$

**Step 3: Deduce the bound for correlation.**

The correlation coefficient is

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}.$$

Taking absolute values,

$$|\rho_{XY}| = \frac{|\text{Cov}(X, Y)|}{\sqrt{\text{Var}(X)\text{Var}(Y)}} \leq 1.$$

Therefore

$$\boxed{-1 \leq \rho_{XY} \leq 1.}$$

**Solution to Problem 10****Step 1: Write the variance of the sample mean.**

For i.i.d. observations  $X_1, \dots, X_n$  with variance  $\sigma^2$ ,

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

We know

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

**Step 2: Observe what happens as  $n$  increases.**

As  $n$  becomes larger,

$$\frac{\sigma^2}{n}$$

becomes smaller.

So the sample mean fluctuates less from sample to sample.

**Step 3: Interpretation in ML and data science.**

A smaller variance of  $\bar{X}$  means:

- empirical averages become more stable,
- estimates of accuracy, loss, or mean feature values become more reliable,
- evaluation metrics vary less due to random sampling noise.

**Step 4: Practical implication.**

With larger datasets:

- training statistics are less noisy,
- validation estimates are more trustworthy,
- model comparisons become more reliable.

Thus

$$\boxed{\text{Var}(\bar{X}) = \frac{\sigma^2}{n} \text{ decreases as } n \text{ increases, which improves statistical stability.}}$$